

Group Project 5: Cost-Aware Harness Optimization

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1. Background

An agent harness is the software layer wrapped around a language model: it renders the prompt, exposes the tool interface, truncates observations, tracks the remaining budget, and decides when the agent stops. TUA-Bench (Meta AI, 2026) evaluates such systems on 120 real-world terminal tasks, each executed in a container and graded by an execution-based verifier. The strongest reported configuration, Claude Code with Claude Opus 4.8, solves only 65.8% of the tasks at roughly \$173.61 per run, while open-weight models under a lightweight scaffold reach 47–48% at \$12–23 per run.

The benchmark's own ablations show that the harness matters as much as the model: changing only the scaffold reverses the ranking of frontier models, and raising the per-task time limit from 150 to 2400 seconds lifts one fixed agent-model pair from 33.0% to 60.1%. Ranking systems by accuracy alone therefore rewards spending rather than engineering. This project asks you to improve the harness under a fixed cost budget and evaluates you on the accuracy–cost trade-off, so that a cheap model with a well-engineered harness can compete with an expensive one.

2. Problem Description

Design and implement harness-level improvements that maximize the task success rate of a fixed base model on TUA-Bench under a fixed per-task cost budget. The optimization surface is the harness only:

- **Context management:** how observations are truncated and what is carried across steps.
- **Tool interface:** tool granularity (one shell tool versus separate read, write, edit and search tools), argument validation, and how errors are reported back to the model.
- **Budget allocation:** the per-task time limit, the maximum number of steps, and whether the remaining budget is shown to the agent.
- **Failure detection and recovery:** detecting repeated or stalled actions, the retry policy, and how the agent recovers after an error.
- **Pre-submission verification:** having the agent check its own artifacts against the task statement before it stops.

The base model, its decoding settings, and the benchmark tasks and verifiers must be held fixed across all compared harness variants; every reported improvement must come from the harness alone. Fine-tuning, task-specific hard-coding and modifying the official verifiers are not allowed.

The central question is whether harness engineering can substitute for model spending: can a cheap base model running your harness reach a higher success rate than an expensive model running the default harness, at a lower cost per run?

3. Evaluation

3.1 Metrics

- **Success Rate:** the mean per-task reward, as defined by the benchmark.
- **Cost per run:** US dollars and tokens per run, including the calls your harness makes itself (summarization, verification, routing).
- **Main metric:** the best success rate reached at or below a given cost per run. Plot your configurations in the accuracy–cost plane; accuracy alone is not the ranking criterion.

3.2 Benchmark and Protocol

- **Benchmark:** TUA-Bench (120 tasks, Harbor orchestration, execution-based verifiers), with the official tasks and verifiers unmodified. A fixed, documented subset of at least 40 tasks is acceptable if compute is limited, as long as every configuration uses the same subset.
- **Cost accounting:** access all models through one provider (for example OpenRouter) so that dollar costs are comparable, and report the price table you used.
- **Repeated runs:** three runs per configuration with the mean reported is preferred; a single run is acceptable if compute is limited, provided you say so.
- **Budget:** report your per-task time and step limits. A variant that improves the success rate only by spending a larger budget should be reported as such.

4. Deliverables

Please submit the following deliverables via a Github repo or a Google Drive folder.

4.1 Codebase

- **A harness under your control:** written from scratch or forked from an open-source one (for example Terminus-2 or mini-swe-agent), wired to the official benchmark runner.
- **At least one improved harness variant** of your own design over the optimization surface in Section 2.
- **Two baselines for comparison:**
 - **Default-harness baseline:** the unmodified harness with its default settings and the same base model.
 - **Expensive-model baseline:** the unmodified harness with a stronger, more expensive base model, at a comparable or higher cost per run.
- **Reproduction material:** configuration files, run commands and the raw result logs behind every reported number.

4.2 Report

The report must contain, at minimum:

- **Accuracy–cost plot:** success rate against dollars per run for every configuration you ran, with the Pareto frontier marked.
- **Component ablation:** a table isolating each harness change, one component at a time from the same baseline.
- **Cost breakdown:** tokens and dollars per run, split into agent inference and harness overhead.
- **Failure classification:** categorize failures into at least two types, with at least one example trajectory for each:
 - **Harness-level failures:** premature termination, truncated or lost context, unparsable tool calls, or no retry after a recoverable error.
 - **Model-level failures:** a wrong plan, misuse of a domain-specific tool, or a task beyond the capability of the base model.
- **Discussion:** which harness components account for most of the gain, and whether that gain shrinks as the base model gets stronger.

4.3 Presentation

Each group will give a **10-minute presentation**, followed by **5 minutes of Q&A**. The presentation should cover:

- The main idea and design of your harness changes.
- Key implementation choices and how your harness is wired to the benchmark runner.
- The most important experimental results, including the comparison with the two baselines in the accuracy–cost plane.
- Representative failure cases and what they say about harness-level versus model-level limits.
- The main conclusions and lessons learned from your project.